

- Motor unit, **1130f**
 Mountain lions, 1149
 Mountain pine beetles (*Dendroctonus ponderosae*), 1244f, 1245, 1280f
 Mountains, 1168, 1169f
 Mount Kilimanjaro, 1189
 Mount St. Helens, 636f, 637
 Mouse (mice), 22
 agouti gene and, 372f
 appetite regulation in, 918
 brains of, 1086f
 comparing human genome with genome of, 454, 455f, 460f–462f
 complete genome sequence for, 454, 455f
 density-dependent population regulation of, 1204f
 energy budgets of, 892
 FOXP2 gene evolution in, 461, 462f
 genomic imprinting of insulin-like growth factor gene of, 310f, 311
 homeotic genes in, 463f
 as model organisms. *See Mus musculus*
 modes of natural selection in, 498f
 osmotic homeostasis in desert, 981
 paw development of, 231f
 transfer of genetic trait between bacterial strains in, 315f
 Mouse (*Peromyscus polionotus*), camouflage case studies with, 2f, 20f–21f, 23
 Mouse model organism, 22
 Mouth formation, 1050f
 Movement, prokaryotic, 576f, 577
 Movement corridors, **1271f**, 1272
 MPF (maturation-promoting factor), **245**, 246f
mPGES-1 gene, 376
 MRSA, 214, 478f
 Mucoromycetes, 660f, **663f**
 Mucous cells, 907f
 Mucus, **906**, 907f, 954
 Mucus escalator, 943
 Mule deer, 871, 1149
 Mules, 509f
 Muller, Hermann, 360
 Müllerian mimicry, 1218f, **1219**
 Multicellular asexual reproduction, 255f
 Multicellular organisms, 5f, 93, 533f, 535f–536f, 675f–676f
 Multienzyme complexes, 161
 Multifactorial characters, **282f**
 Multifactorial disorders, human, 287
 Multigene families, 452, **453f**
 Multiple fruits, **831f**
 Multiple myeloma, 65
 Multiple sclerosis, 971
 Multiplication rule, **277f**, 278
 Multiprotein complexes, 174f, 175
 Mumps, 409
 Murchison meteorite, 527
Muscardinus avellanarius, 893f
 Muscle
 cardiac and smooth, 1131–1132
 contraction of, 1126, 1127f, 1128
 regulation of contraction of, 1128, 1129f–1130f
 skeletal. *See* Skeletal muscle
 Muscle cells, 117f, 383, 384f, 673, 915
 Muscle fibers, 879f, 1130, 1131t
 Muscle tissue, **879f**
 Muscular dystrophy, 299, 433
 Mushrooms, 654f, 655, 665f–667f, 670. *See also* Fungi
Mus musculus (mouse), 22. *See also* Mouse
 Mussels, 505, 701f, 702f, 1264
 Mustard plant model organism. *See Arabidopsis thaliana*
 Mutagens, **360**
 Mutant phenotypes, 295f–296f
 Mutants
 designing experiments using genetic, 1095
 interpreting data from experiments with genetic, 918
 in molecular biology, 845
 nutritional, in gene-enzyme relationship experiment, 336f–338f
 Mutations, **357f**
 in alterations of chromosome structure, 454f–455f
 in aquaporins causing diabetes insipidus, 995f
 in cancer development, 391f, 394
 cancer genes and, 388, 389f
 cellular slime mold, 613
 CRISPR-Cas9 and gene editing for, 426, 434–435
 of developmental genes, 545f–546f
 in duplication and divergence of gene-sized regions, 455f–457f
 in duplication of entire chromosome sets, 454
 effects of, during cell division, 389f–390f
 as embryonic lethals, 386
 as errors in proofreading, 327
 evolution and rate of, 334
 evolution of enzymes by, 159f
 in exon duplication and exon shuffling, 456, 457f
 in flowering, 779f–780f
 gene editing to correct, 360, 361f
 genetic variation from, 305f–306f
 genome evolution and, 454f–457f, 458–459
 Hardy-Weinberg equilibrium and, 492t
 of ion channel protein genes, 1076
 in mitochondrial DNA, 311
 molecular clock speed and, 566, 567f
 mutagens as cause of, 360
 natural selection and, 328
 nucleotide-pair substitutions, insertions, and deletions, 357, 358f, 359
 phenotypes and, 295f–296f, 306f
 plants and creation of, in molecular biology, 776
 point mutations, 357f–358f, 359
 in prokaryotes, 573, 578–579
 as source of alleles, 265
 as sources of genetic variation, 488–489
 transposable elements and, 459
 of viruses, 409, 414
 Mutualism, **587**, **1220**, 1256f–1257f
 bacterial, 587f
 in flower pollination, 825f
 fungal, 655, 661f, 667f–669f. *See also* Mycorrhizae
 as interspecific interaction, 1220f, 1221
 across kingdoms and domains, 813f
 mycorrhizae as plant-fungi, 787, 817, 818f
 nutrient limitations and, 1244
 plant-bacteria, 813f–816f, 817
 in vertebrate digestive systems, 912, 913f–914f
 Myasthenia gravis, 1129
 Mycelium (mycelia), 655f–659f, **656**, 662f
 Mycetozoans, 612f–613f
Mycobacterium tuberculosis, 589, 957
 Mycorrhizae, **656**, **787**, **817**
 basidiomycetes in, 665
 biological augmentation using, 1255
 in colonization of land by plants, 536, 537f
 evolution of, 660
 genomic analysis of interactions of, 657
 as mutualism, 1220–1221
 nutrient limitations and, 1244
 plant nutrition and, 817, 818f
 plant roots and, 760
 as root-fungi mutualism, 787
 specialized hyphae in, 656f, 657
 strigolactones and, 855
 terrestrial plants and, 621
 Mycorrhizal associations, 760
 Mycosis, **670**
 Myelination, 1076f–1077f
 Myelin sheath, **1076f**–1077f
 Myeloid stem cells, 935f
Myllokunmingia fengjiao, 718f, 724
 Myoblasts, 383, 384f
 Myocardial infarctions, 937f, 938–939
 MyoD activator, 374f
myoD gene, 383, 384f
 Myofibrils, **1126f**
 Myoglobin, **949**, **1130**, 1131t
 Myopathy, 311
 Myosin, 76f, **117**, 241, 384f, 879f
 Myosin filaments, 1125, 1126f–1127f
 Myotonia, 1034, 1076
 Myriapods, **707**–708, 709f
Myrmecocystus, 485
Mytilus edulis, 505
 Myxini (hagfishes), 719f, 723f
 Myxobacteria, 213f
Myxococcus xanthus, 213f
 NADH, 172f–173f, 178, 180f, 181
 NADP⁺ (nicotinamide adenine dinucleotide phosphate), **191f**, 192, 206, 207f
 NADPH, **191f**, 192, 197f–198f, 206, 207f
 Naked mole rats, 1157f, 1158
 Naloxone, 1082
 Nanopores, 416
 Nasal glands, marine bird, 982f
 Nash, John, 1154
 National Cancer Institute, 447
 National Center for Biotechnology Information (NCBI), 444, 445f
 National Institutes of Health (NIH), 444, 447
 National Library of Medicine (NLM), 444
 National Medal of Science, 944
 Native Americans, 565
 Natural family planning, 1037, 1038f
 Natural killer (NK) cells, **955**
 Natural logarithms in Scientific Skills Exercise, 639
 Natural plastics, 590, 591f
 Natural range expansions, 1183, 1184f
 Natural selection, **14**, **473**. *See also* Adaptations; Evolution
 adaptations and, 472, 473f
 adaptive evolution and, 494
 Darwinian theory of descent with modification by, 14f–16f, 469–470, 473f–476f, 483–484
 of developmental genes, 545
 directional, disruptive, and stabilizing selection in, 497, 498f
 of ecological niches, 1215, 1216f
 evolution of drug resistance by, 478f
 evolution of enzymes by, 159f
 in evolution of life history traits, 1200, 1201f–1202f
 genetic variation for, from genetic recombination, 305
 Hardy-Weinberg equilibrium and, 492t
 insect evolution by, due to food source changes, 477f, 478
 key role of, in adaptive evolution, 498, 499f
 limitations of, in creating perfect organisms, 501f, 504
 in macroevolution, 551
 mutations and, 328
 predator role in, 26
 relative fitness and, 497
 of ribozymes, 528
 sexual reproduction, genetic variation, and, 266, 267f
 species selection as, 548–549
 Natural vs. supernatural explanations, 18
 Nature reserves
 philosophy of, 1272–1273
 zoned reserves and, 1273f
 Nature vs. nurture, 282f, 762
 Nautilus, chambered, 701f, 702
 ncRNAs, 379, 380f, 381
 Neanderthals (*Homo neanderthalensis*), 33, 88f, 461, 752f–753f
 Near vision, 1122f
 Nectar guides, 824f
 Nectaries, 813f
 Nectarine, 645f
 Negative and positive correlations, 834
 Negative feedback, **882**, **1005**
 in density-dependent population growth, 1203
 in endocrine system feedback regulation, 1005–1006
 in feedback regulation, 10f
 in homeostasis, 882
 Negative gene regulation, bacterial, 369
 Negative gravitropism, 861
 Negative pressure breathing, **945f**, 946
Neisseria gonorrhoeae, 576, 584f
 Nematocysts, **691**, 692f
 Nematode model organism. *See Caenorhabditis elegans*
 Nematodes, 689f, 705f, 706, 1058f–1059f, 1133f
 Nemertea, 688f
Nemoria arizonaria, 488f
Neodenticula seminiae (diatom), 1170
 Neon, 35f
 Neonatal diabetes, 359
 Neoproterozoic era, 530f, 676f–677f
 Neornithes, 739
 NEP. *See* Net ecosystem production
 Nephrons, **986f**